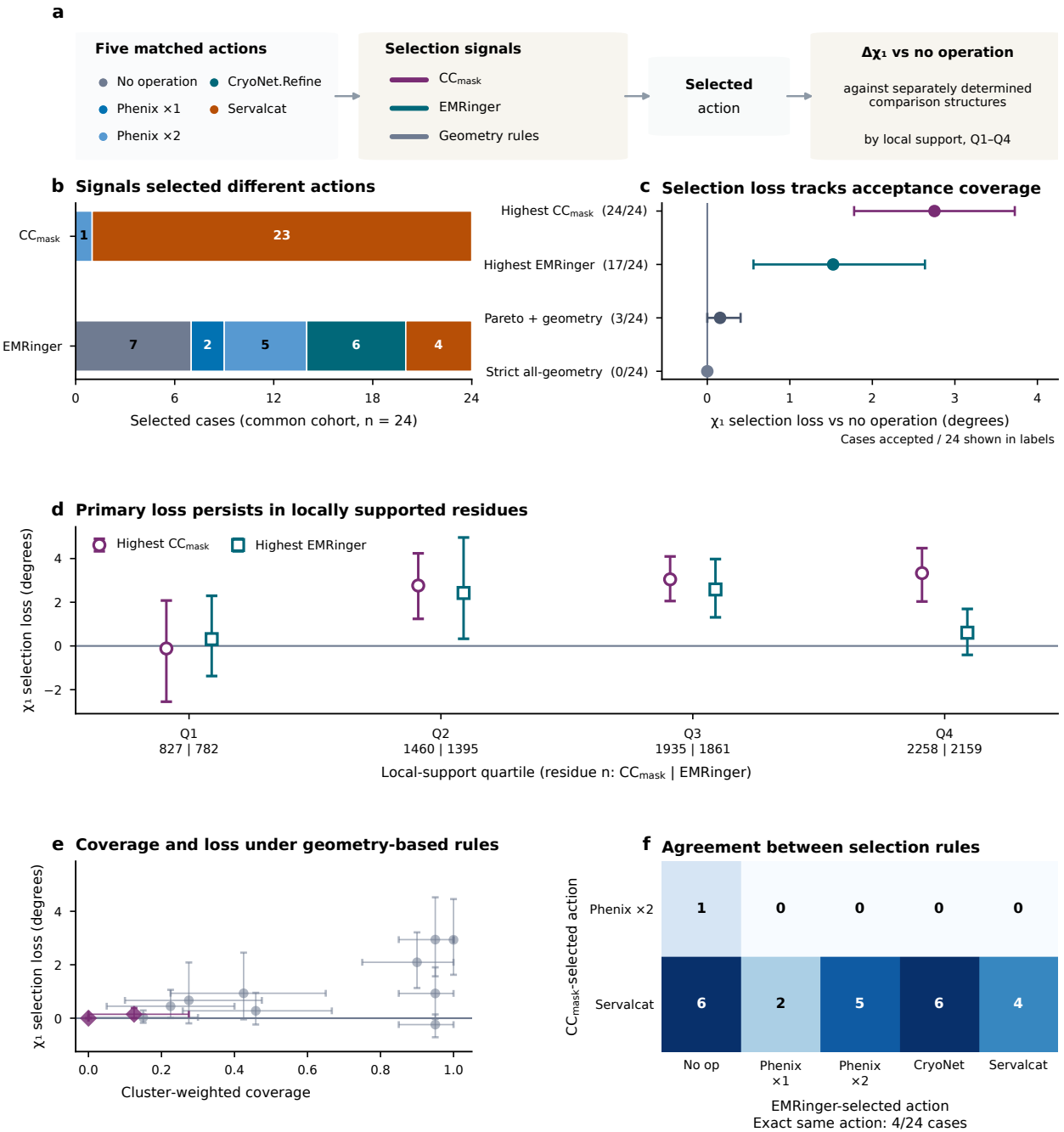

a
Five matched actions
No operation
CryoNet.Refine
Phenix ×1
Servalcat
Phenix ×2
Selection signals
CCmask
EMRinger
Geometry rules
Selected action
Δχ1 vs no operation
against separately determined comparison structures
by local support, Q1–Q4
b Signals selected different actions
CCmask
EMRinger
1
23
7
2
5
6
4
0
6
12
18
24
Selected cases (common cohort, n = 24)
c Selection loss tracks acceptance coverage
Highest CCmask (24/24)
Highest EMRinger (17/24)
Pareto + geometry (3/24)
Strict all-geometry (0/24)
0
1
2
3
4
χ1 selection loss vs no operation (degrees)
Cases accepted / 24 shown in labels
d Primary loss persists in locally supported residues
Highest CCmask
Highest EMRinger
χ1 selection loss (degrees)
4
2
0
−2
Q1
827 | 782
Q2
1460 | 1395
Q3
1935 | 1861
Q4
2258 | 2159
Local-support quartile (residue n: CCmask | EMRinger)
e Coverage and loss under geometry-based rules
χ1 selection loss (degrees)
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
Cluster-weighted coverage
f Agreement between selection rules
CCmask-selected action
Phenix ×2
Servalcat
1
0
0
0
0
6
2
5
6
4
No op
Phenix ×1
Phenix ×2
CryoNet
Servalcat
EMRinger-selected action
Exact same action: 4/24 cases